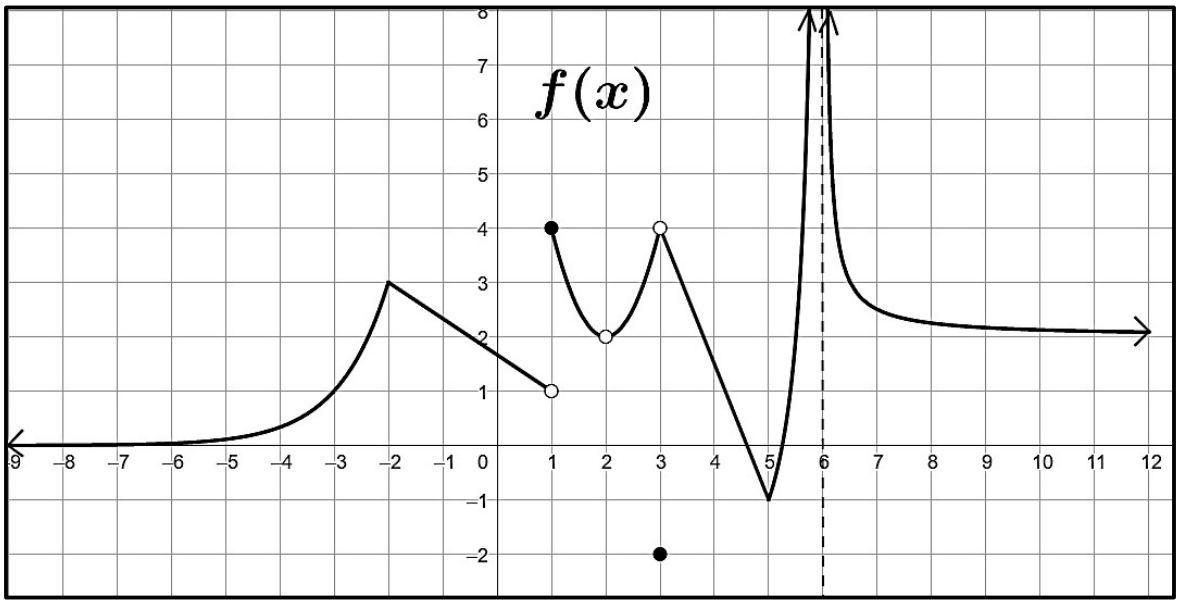
Worksheet A Review of Unit 1
AP Calculus Name_____



Directions: For 1– 9, use the graph of $f(x)$ above to answer the following problems.

1. $\lim_{x\to 3} f(x)$
2. $\lim_{x\to 6} f(x)$
3. $\lim_{x\to 2} f(x)$
4. $\lim_{x\to 1^-} f(x)$
5. $f(3)$
6. $\lim_{x\to -\infty} f(x)$

7. Which of the following statements are true?

I. $\lim_{x\to 1^-} f(x) = \lim_{x\to 1^+} f(x)$ II. $\lim_{x\to 3} f(x) = \lim_{x\to 1^+} f(x)$ III. $\lim_{x\to \infty} f(x) = \lim_{x\to 2} f(x)$

A) I only B) II only C) III only D) I and II only
E) I and III only F) II and III only G) I, II, and III

8. For which value(s) of a does $\lim_{x\to a} f(x) = 1$?

9. For which value(s) of x does $f(x)$ has a removable discontinuity?

Directions: Find the following limits if possible. Be sure to show all work and proper notation.

10. $\lim_{x\to -2} \dfrac{x^2 - 5x - 14}{x^2 + 6x + 8}$

11. $\lim_{x\to \infty} \dfrac{(2x-3)^2(5-3x)}{4x^3 - 3x + 5}$

12. $\lim_{x \to 0} \frac{2x^2 - 5x^3}{6x^5 - x^2}$

13. $\lim_{x \to 3} \frac{x^2 - 9}{x^2 - 6x + 9}$

14. $\lim_{x \to \infty} \frac{4 - 7x}{\sqrt{9x^2 + 2x - 7}}$

15. $\lim_{x \to -5} \frac{x^2 + 8x + 15}{4x + 20}$

16. $\lim_{x \to 2} \frac{|x - 2|}{3(x - 2)}$

17. $\lim_{x \to -\infty} \frac{\sqrt{4x^2 + 7}}{5x - 1}$

18. $\lim_{x \to -1^-} \frac{2x + 2}{|x + 1|}$

19. $\lim_{x \to 0} \frac{4x^3 + 7x}{3x^2 + 2x}$

20. $\lim_{x \to -\infty} \frac{3 + 4^x}{3^x - 2}$

21. $\lim_{x \to \infty} \frac{x^{99}}{3^x}$

$$f(x) = \begin{cases} 3x^2 - a, & x < 1 \\ 2x + 5a, & x \geq 1 \end{cases}$$

22. Find the value of a that makes $f(x)$ continuous.

$$g(x) = \begin{cases} \dfrac{x^2 + 2x - 15}{x^2 - 10x + 21}, & x \neq 3 \\ k, & x = 3 \end{cases}$$

23. Find the value of k that makes $g(x)$ continuous.

24. Find any horizontal asymptotes for the following functions.

a. $k(x) = \frac{x^2 - 3x + 5}{4x^2 - 6}$

b. $m(x) = \frac{4^x - 5}{10 + 4^x}$

c. $p(x) = \dfrac{\sqrt{4x^2 + 7}}{6 - 8x}$

$$f(x) = \begin{cases} 3x - 11, & x < 4 \\ x^2 - 5x + 5, & x \geq 4 \end{cases}$$

25. Is $f(x)$ continuous at $x = 4$? Give a reason for your answer.

x	1	3	4	7
$g(x)$	4	7	2	0

26. The function $g(x)$ is continuous for all real values with selected values included in the table above. Is there a time c, $1 < c < 7$, such that $g(x) = 1$? Give a reason for your answer.